2.2 Minimization techniques of Logical equations

About K map

- A Karnaugh map (K-map for short) is a useful graphical tool used in the simplification of combinational boolean equations and the creation of sequential logic circuits. Karnaugh maps were created by Maurice Karnaugh in 1953.
- It contains boxes which are filled according to information contained in the truth table.
- K-map is ideally suited for designing the combinational logic circuits using either a SOP or POS method.

- The philosophy behind these drawings is that differences of only one bit for the logic of a boolean equation are adjacent to each other.
- This is just an organizational method for a boolean logic truth table, but it can give you the ability to help simplify logical equations.
- This has proven to be especially useful for digital circuit designers, as it can suggest components which can be eliminated or a way to simplify circuit designs.
- This reduces both the cost and complexity of these designs, and even an automated method for developing these circuits assuming that you can come up with a logical truth table in the first place.
- What is going to be demonstrated here is how to manually evaluate Karnaugh Maps.
- For very complex circuit designs that involve dozens or even hundreds of variables, there is software available that automates this process.

Structure and Creation

- The structure of a Karnaugh map is grid shaped. The two most typical sizes used for instruction or for small projects is the three variable (a 2x4 grid or 4x2 depending on the user) and the four variable map (4x4 grid).
- K-maps can be written for 2,3,4.. up to 6 variables as shown below. Beyond that K-map technique becomes very cumbersome.

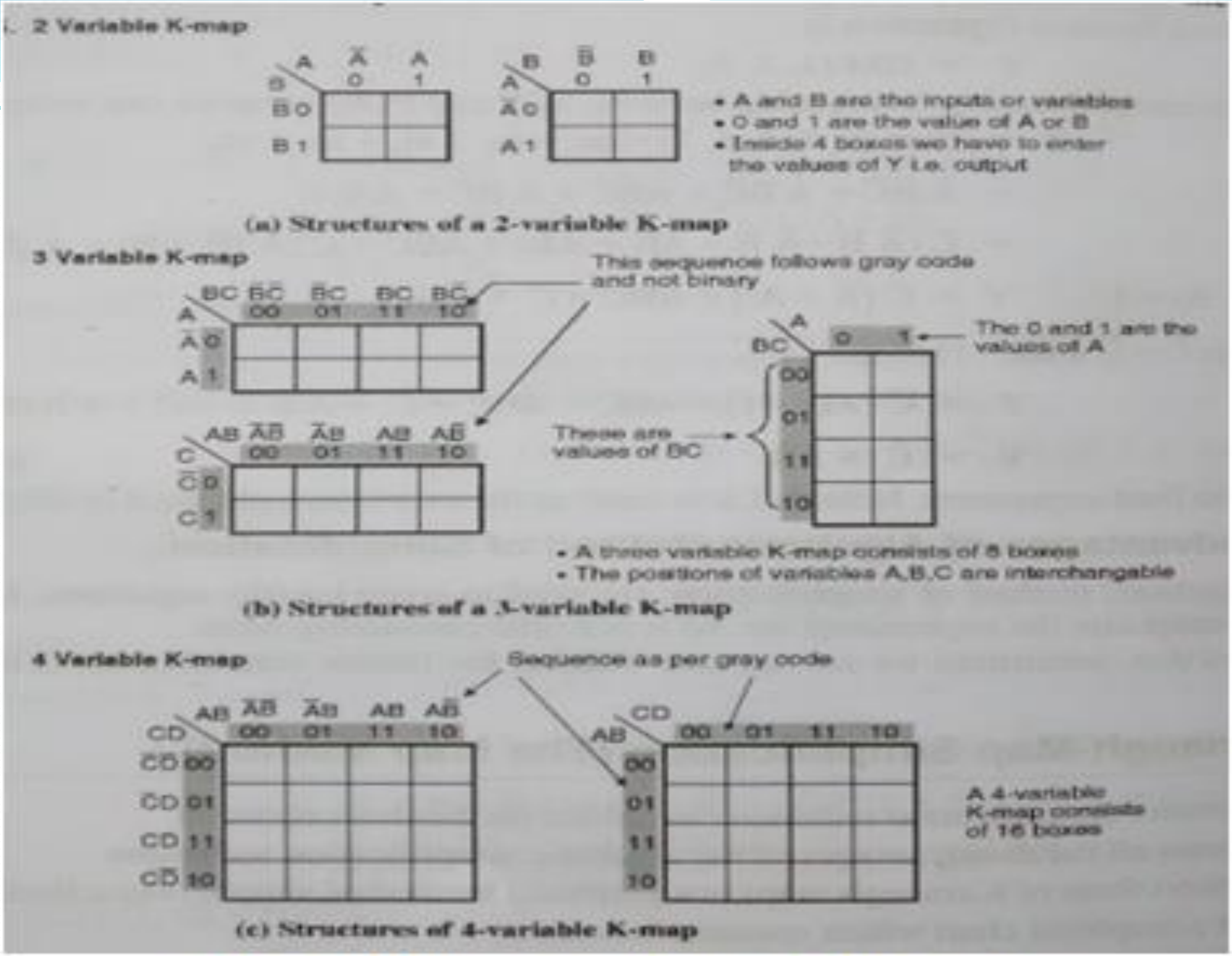

Fig1. Structure of 2, 3 and 4 variable K-map

Structure of 2 variable K map

A. SOP: -

A \ B	$\overline{B}$ 0	B 1
$\overline{A}$ 0	$\overline{A}.\overline{B}$	$\overline{A}.B$
A 1	$A.\overline{B}$	$A.B$

B. POS: -

A \ B	B 0	$\overline{B}$ 1
A 0	$A+B$	$A+\overline{B}$
$\overline{A}$ 1	$\overline{A}+B$	$\overline{A}+\overline{B}$

Structure of 3 variable K map

A \ B,C	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

A. SOP: -

A \ BC	$\overline{B}\,\overline{C}$ 0 0	$\overline{B}\,C$ 0 1	$B\,C$ 1 1	$B\,\overline{C}$ 1 0
$\overline{A}$ 0	$\overline{A}\,\overline{B}\,\overline{C}$ 0	$\overline{A}\,\overline{B}\,C$ 1	$\overline{A}\,B\,C$ 3	$\overline{A}\,B\,\overline{C}$ 2
A 1	$A\,\overline{B}\,\overline{C}$ 4	$A\,\overline{B}\,C$ 5	$A\,B\,C$ 7	$A\,B\,\overline{C}$ 6

AB \ C	$\overline{C}$ 0	C 1
$\overline{A}\,\overline{B}$ 00	$\overline{A}\,\overline{B}\,\overline{C}$ 0	$\overline{A}\,\overline{B}\,C$ 1
$\overline{A}\,B$ 01	$\overline{A}\,B\,\overline{C}$ 2	$\overline{A}\,B\,C$ 3
A B 11	$A\,B\,\overline{C}$ 6	$A\,B\,C$ 7
$A\,\overline{B}$ 10	$A\,\overline{B}\,\overline{C}$ 4	$A\,\overline{B}\,C$ 5

B. POS: -

A \ B+C	B+C 0 0	$B+\overline{C}$ 0 1	$\overline{B}+\overline{C}$ 1 1	$\overline{B}+C$ 1 0
A 0	A+B+C 0	$A+B+\overline{C}$ 1	$A+\overline{B}+\overline{C}$ 3	$A+\overline{B}+C$ 2
$\overline{A}$ 1	$\overline{A}+B+C$ 4	$\overline{A}+B+\overline{C}$ 5	$\overline{A}+\overline{B}+\overline{C}$ 7	$\overline{A}+\overline{B}+C$ 6

A+B \ C	C 0	$\overline{C}$ 1
A+B 0 0	A+B+C 0	$A+B+\overline{C}$ 1
$A+\overline{B}$ 0 1	$A+\overline{B}+C$ 2	$A+\overline{B}+\overline{C}$ 3
$\overline{A}+\overline{B}$ 1 1	$\overline{A}+\overline{B}+C$ 6	$\overline{A}+\overline{B}+\overline{C}$ 7
$\overline{A}+B$ 1 0	$\overline{A}+B+C$ 4	$\overline{A}+B+\overline{C}$ 5

Structure of 4 variable K map

CD \ AB	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

ABCD	ABCD
0000 - 0	1000 - 8
0001 - 1	1001 - 9
0010 - 2	1010 - 10
0011 - 3	1011 - 11
0100 - 4	1100 - 12
0101 - 5	1101 - 13
0110 - 6	1110 - 14
0111 - 7	1111 - 15

A,B \ C,D	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

A. **SOP: -**

$AB \backslash CD$	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	$\bar{A}\bar{B}\bar{C}\bar{D}$ 0	$\bar{A}\bar{B}\bar{C}D$ 1	$\bar{A}\bar{B}CD$ 3	$\bar{A}\bar{B}C\bar{D}$ 2
$\bar{A}B$ 01	$\bar{A}B\bar{C}\bar{D}$ 4	$\bar{A}B\bar{C}D$ 5	$\bar{A}BCD$ 7	$\bar{A}BC\bar{D}$ 6
AB 11	$AB\bar{C}\bar{D}$ 12	$AB\bar{C}D$ 13	$ABCD$ 15	$ABC\bar{D}$ 14
$A\bar{B}$ 10	$A\bar{B}\bar{C}\bar{D}$ 8	$A\bar{B}\bar{C}D$ 9	$A\bar{B}CD$ 11	$A\bar{B}C\bar{D}$ 10

B. **POS: -**

$A+B \backslash C+D$	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+\bar{D}$ 11	$\bar{C}+D$ 10
$A+B$ 00	$A+B+C+D$ 0	$A+B+C+\bar{D}$ 1	$A+B+\bar{C}+\bar{D}$ 3	$A+B+\bar{C}+D$ 2
$A+\bar{B}$ 01	$A+\bar{B}+C+D$ 4	$A+\bar{B}+C+\bar{D}$ 5	$A+\bar{B}+\bar{C}+\bar{D}$ 7	$A+\bar{B}+\bar{C}+D$ 6
$\bar{A}+\bar{B}$ 11	$\bar{A}+\bar{B}+C+D$ 12	$\bar{A}+\bar{B}+C+\bar{D}$ 13	$\bar{A}+\bar{B}+\bar{C}+\bar{D}$ 15	$\bar{A}+\bar{B}+\bar{C}+D$ 14
$\bar{A}+B$ 10	$\bar{A}+B+C+D$ 8	$\bar{A}+B+C+\bar{D}$ 9	$\bar{A}+B+\bar{C}+\bar{D}$ 11	$\bar{A}+B+\bar{C}+D$ 10

- K-maps can only be created if a truth table is present; it works differently for sequential logic
- A finished K-map can make a truth table, or a boolean equation vice versa.
- Once a truth table is acquired, a Karnaugh map can be created. The top left corner(or sometimes just the top and the left) of a K-map shows the variables used for that side.
- The picture of the K-map example shows the variables associated to that side.
- The top is A and B and the numbers below them is the state A and B are for that column (i.e. The column 10 is when A is true(high) and B is false(low)). From the truth table we put its output in the corresponding squares (i.e. If on the truth table when ABCD is 1011 and the output was 1, then on the four variable map, a 1 would go in the fourth column, second row).

- The number of boxes in a K-map is decided by the number of variables and is equal to 2n, where n is number of variables. Thus for a 4 variable equation, the K-map will contain 16 boxes.
- **The K-map is filled as per grey code so that as we move along a row or column, only one variable will change its value.**
- The simplification of a boolean expression using K-map involves combining or grouping of two or more adjacent cells so as to form a group of pairs or quads or octets, etc.
- If an expression is to be minimized in SOP form, the K-map is filled with 1s corresponding to the minterms. However, the expression is to be minimized in POS form, the K-map is filled with 0s corresponding to the maxterms. Don't care conditions are marked with an “X” and are grouped if required or else ignored.

Order of input values

- The input values combinations at two adjacent rows or columns must differ in exactly one bit.
- 2. The map is cyclic: the first and last rows are considered to be adjacent, and the first and last columns are also adjacent.
- 3. Each possible combination of bits must appear exactly once in the map.

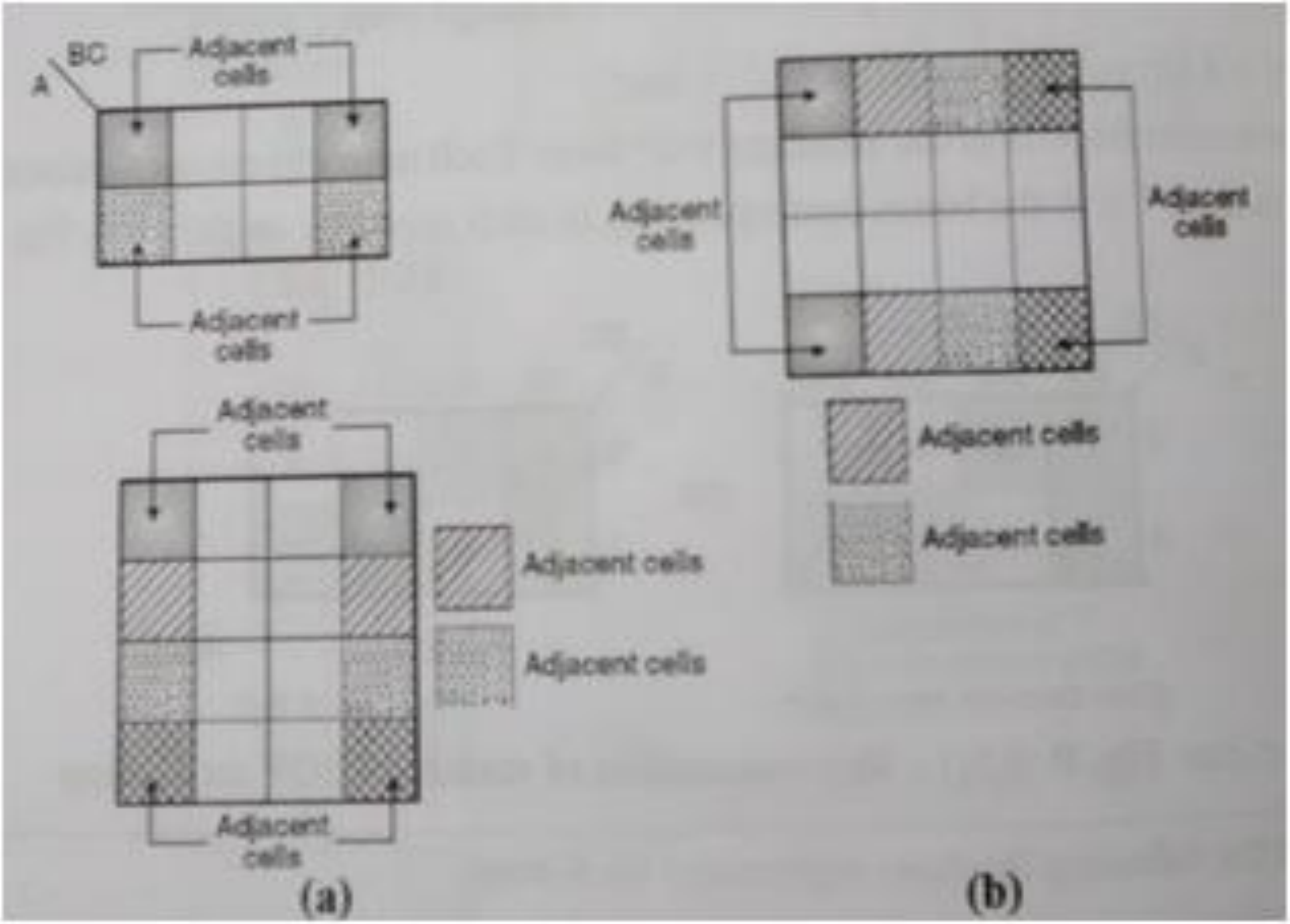

Fig2. Adjacent cells in a K-map

Function Minimization Using K-Maps

1. Each square (minterm) in a k-map of 2 variables has 2 logically adjacent squares, each square in a 3-variable k-map has 3 adjacent squares, etc.

2. Combine only the minterms for which the function is 1.

3. When combining terms on a k-map, group adjacent squares in groups of powers of 2 (I.e. 2, 4, 8, etc.). Grouping two squares eliminates one variables, grouping 4 squares eliminates 2 variables, etc.
   – Can't combine a group of 3 minterms

Function Minimization Using K-Maps

4. Group as many squares together as possible; the larger the group is the fewer the number of literals in the resulting product term

5. Select as few groups as possible to cover all the minterms of the functions. A minterm is **covered** if it is included in at least one group. Each minterm may be covered as many times as it is needed; however, it must be covered at least once.

6. In combining squares on the map, always begin with those squares for which there are the fewest number of adjacent squares (the "loneliest" squares on the map).

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Advantages

- Data representation's simplicity.
- Easy and Convenient to implement.
- Reduces the cost and quantity of logical gates.
- Less number of steps when compared to algebraic minimization technique.
- Does not demand for the knowledge of boolean algebraic theorems.

Disadvantage

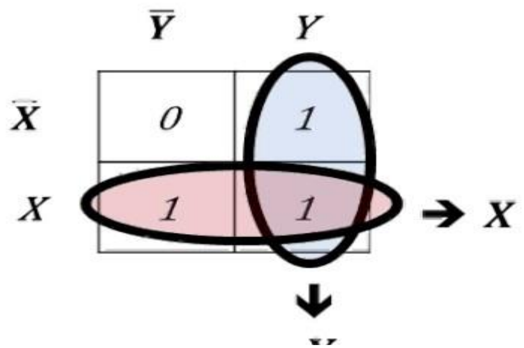
- Complexity of **K-map** simplification process increases with the increase in the number of variables.

Example

F= X’ Y+X Y’ +X Y
 = X’Y + X(Y’+Y)
 = X’Y+ X
 = X + Y

Consider the sample expression

$$\bar{X}.Y + X.\bar{Y} + X.Y$$



F = X + Y

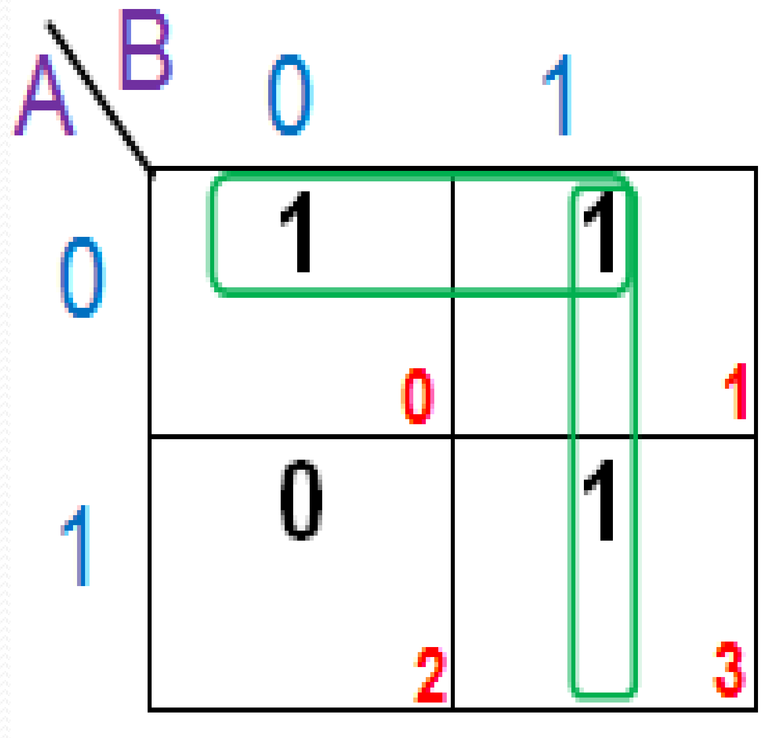
Stage 2: Identify paired loops to simplify

Example

A	B	F
0	0	1
0	1	1
1	0	0
1	1	1

$F = A'B' + A'B + AB$

$F = \sum (0, 1, 3)$



$F = A' + B$

K Map Example

F(a, b, c) = $\sum$m(1,3,5) --SOP

A \ BC	00	01	11	10
0	0	1	1	0
1	0	1	0	0

$$F = B'C + A'C$$

$$= C(B'+A')$$

F(a, b, c)=$\prod$M(0,2,4,6,7)--POS

A \ B+C	00	01	11	10
0	0	1	1	0
1	0	1	0	0

F= (A'+B') .(C)

=

Get a Minimum POS Using K Map

- Cover 0's to get simplified POS.
  - We want 0 in each term.

$$f = x'z' + wyz + w'y'z' + x'y$$

First, the 1's of f are plotted . Then, from the 0's,

$$f' = y'z + wxz' + w'xy$$

and the minimum product of sums for f is

$$f = (y + z')(w' + x' + z)(w + x' + y')$$

$yz \backslash wx$	00	01	11	10
00	1	1	0	1
01	0	0	0	0
11	1	0	1	1
10	1	0	0	1

Three-Variable K-Map

<u>Example 4 (continued)</u> $F(A,B,C) = \Sigma(1,2,3,5,7)$

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

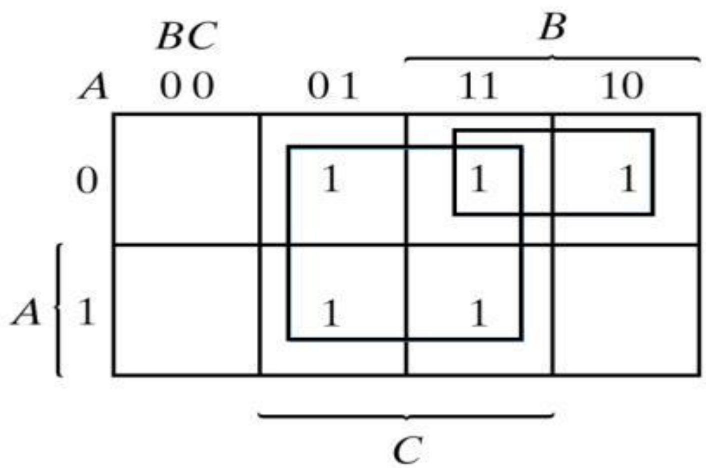


Fig. 3-7 Map for Example 3-4; $A'C + A'B + AB'C + BC = C + A'B$ 10

KARNAUGH MAP MINIMIZATION

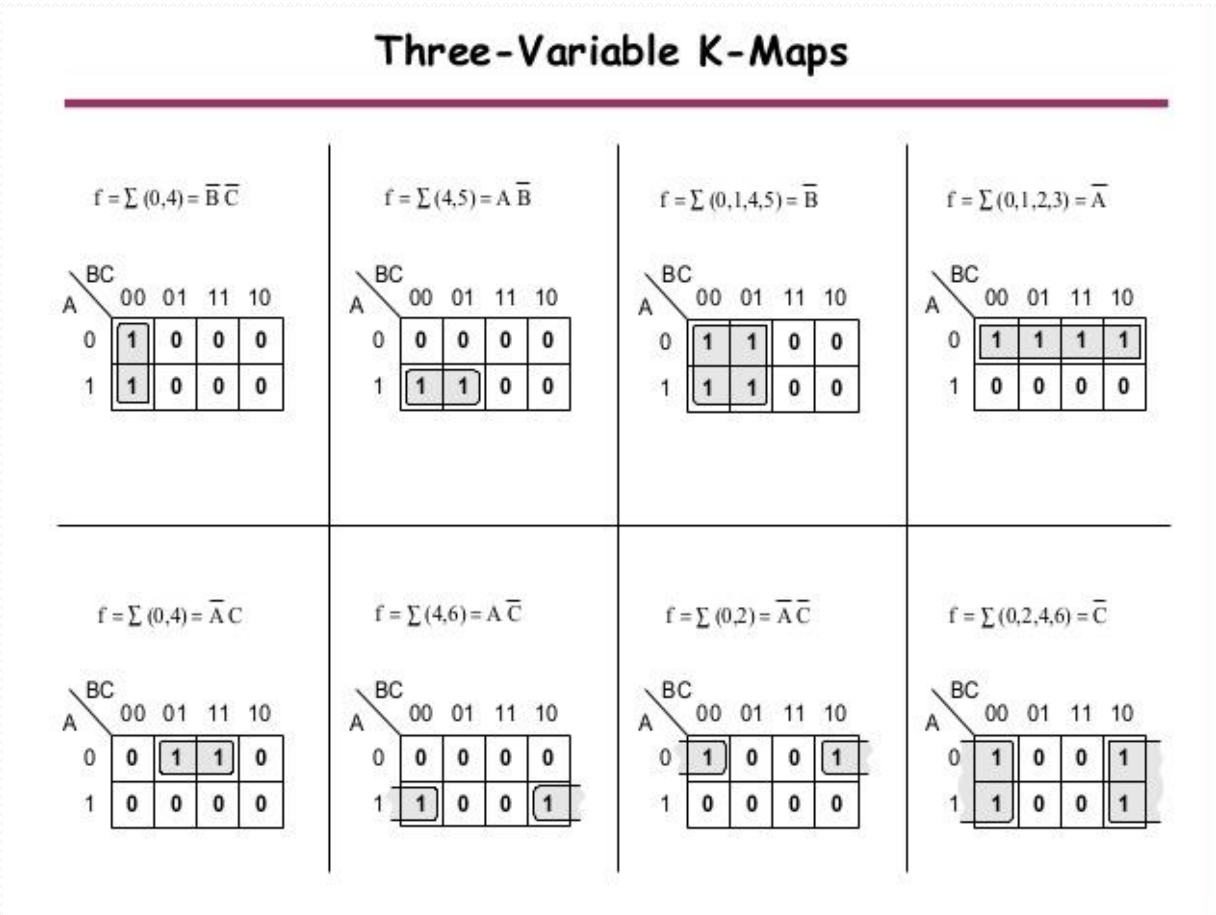
· The leftmost cell/cells can be grouped with the rightmost cell/cells and the topmost cell/cells can be grouped with the bottommost cell/cells.

A \ BC	00	01	11	10
0	1	0	0	1
1	1	0	0	1

Proper grouping

$$F = C'$$

Examples with three variables K map



Four-Variable K-Maps

AB \ CD	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	0	0	0

$f = \sum(0,8) = \overline{B} \bullet \overline{C} \bullet \overline{D}$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	1	0	0
10	0	0	0	0

$f = \sum(5,13) = B \bullet \overline{C} \bullet D$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	1	1	0
10	0	0	0	0

$f = \sum(13,15) = A \bullet B \bullet D$

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	0	0	0	0
10	0	0	0	0

$f = \sum(4,6) = \overline{A} \bullet B \bullet \overline{D}$

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	0	0

$f = \sum(2,3,6,7) = \overline{A} \bullet C$

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$f = \sum(4,6,12,14) = B \bullet \overline{D}$

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	0	0
11	0	0	0	0
10	0	0	1	1

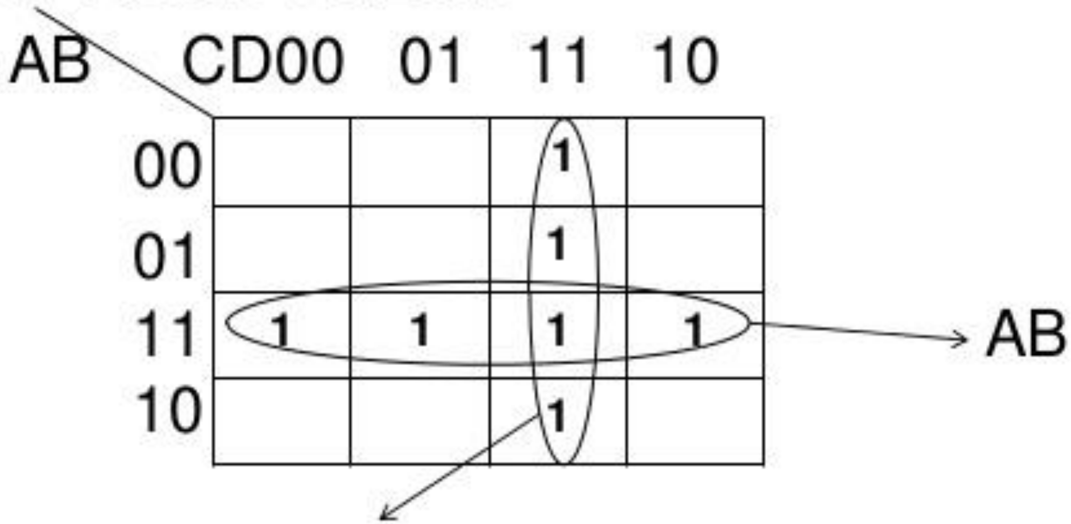
$f = \sum(2,3,10,11) = \overline{B} \bullet C$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$f = \sum(0,2,8,10) = \overline{B} \bullet \overline{D}$

KARNAUGH MAP MINIMIZATION

$$f(A,B,C,D) = \bar{A}\bar{B}CD + \bar{A}BCD + ABCD + A\bar{B}CD +$$
$$AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D}$$



BC

= AB+ C D

Example with don't care case

Don't Care Conditions

- Simplify the Boolean function
  $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ which has the don't-care conditions: $d(w, x, y, z) = (0, 2, 5)$

WX \ YZ	00	01	11	10
00	X	1	1	X
01	0	X	1	0
11	0	0	1	0
10	0	0	1	0

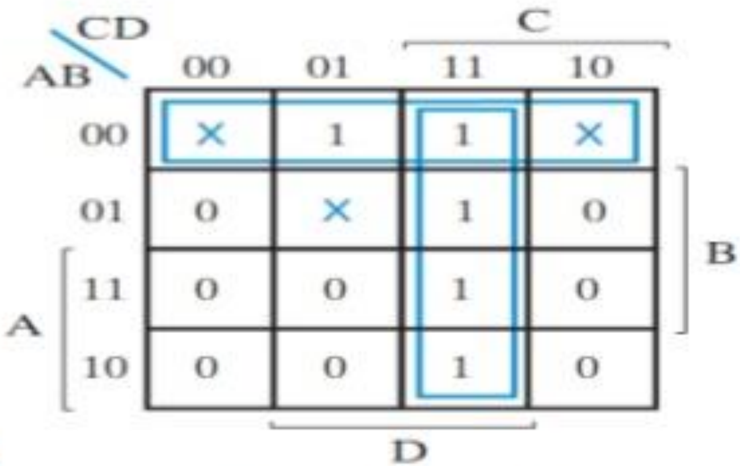
W'X'

YZ

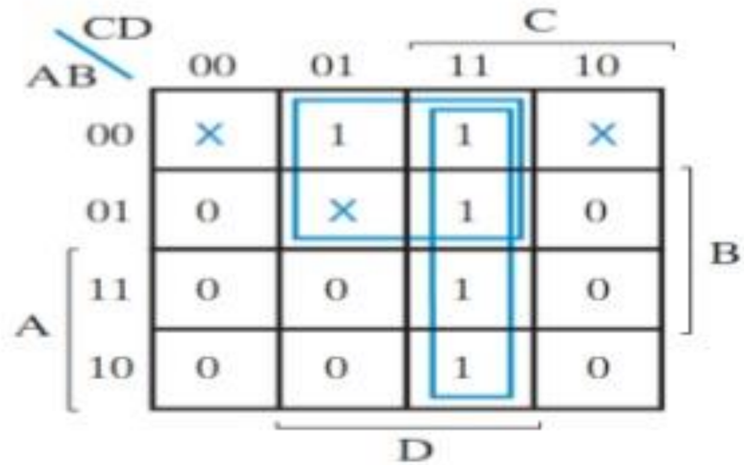
F= W'X' + YZ

EXAMPLES OF DON'T CARE CONDITIONS (2/2)

$F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + \sum d(0, 2, 5)$



(a) $F = CD + \overline{A}\,\overline{B}$



(b) $F = CD + \overline{A}D$

Two possible solutions , both are acceptable.

Five variable k map adjancy

$\bar{A}$

BC \ DE	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	0	1	0
10	1	0	0	0

A

BC \ DE	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	0	1	0
10	1	0	0	0

Groups of 2

$\overline{A}$

BC \ DE	00	01	11	10
00	1	0	0	0
01	0	0	0	1
11	0	0	0	1
10	1	0	0	0

A

BC \ DE	00	01	11	10
00	1	0	0	0
01	0	0	0	1
11	0	0	0	1
10	1	0	0	0

Groups of 4

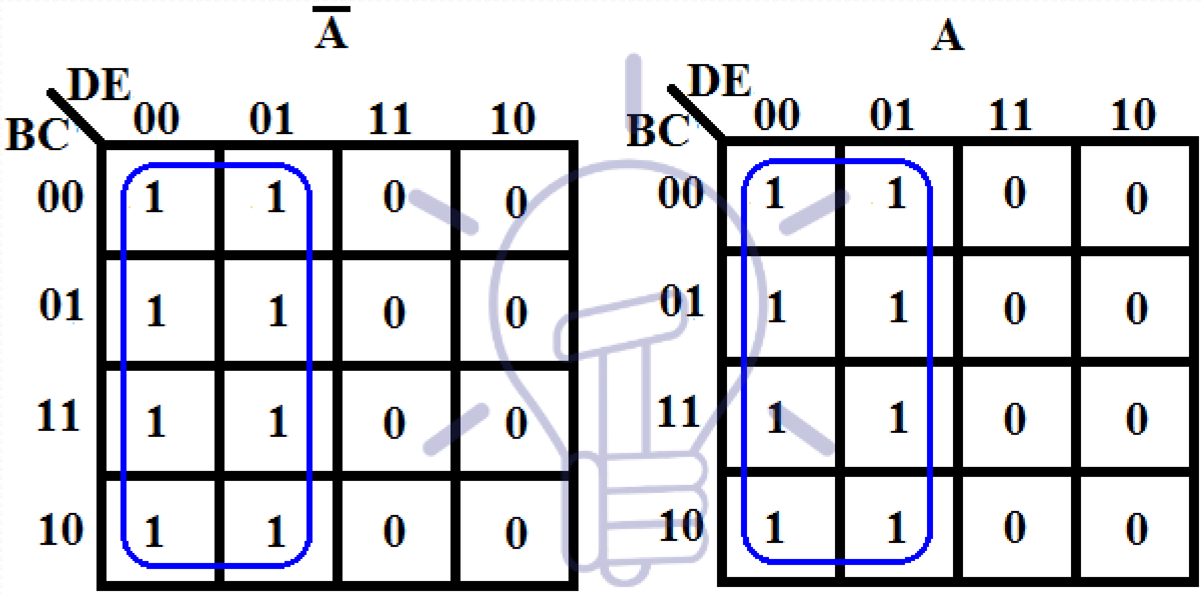
$\overline{A}$

BC \ DE	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

A

BC \ DE	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

Groups of 8


$\bar{A}$
DE
BC
00
01
11
10
00
01
11
10
1 1 0 0
1 1 0 0
1 1 0 0
1 1 0 0
A
DE
BC
00
01
11
10
00
01
11
10
1 1 0 0
1 1 0 0
1 1 0 0
1 1 0 0

Group of 16

Example 1 with 5 variable K-map

- F (A,B,C,D,E) = $\sum$ ($m_0, m_2, m_5, m_7, m_8, m_{10}, m_{16}, m_{21}, m_{23}, m_{24}, m_{27}, m_{31}$)

$\overline{A}$

BC \ DE	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	0	0	0
10	1	0	0	1

A

BC \ DE	00	01	11	10
00	1	0	0	0
01	0	1	1	0
11	0	0	1	0
10	1	0	1	0

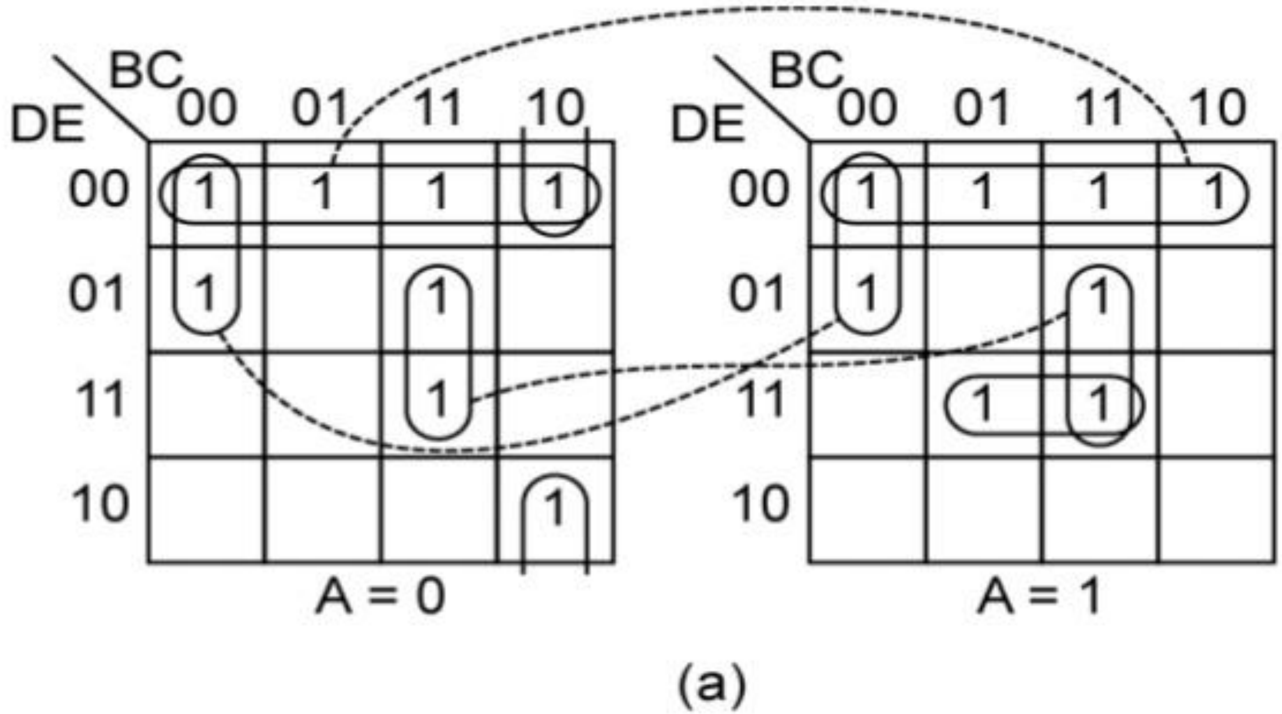
- F= A'C'E' + C'D'E' + B'CE + ABDE

Explanation of 5 variable k map example

- There are four groups made in this K-map. Each group has a different color to differentiate between them.
- The red color group is a group of 4 min terms made between both 4-variable k-maps because they are adjacent cells and it overlaps the green group.
- The yellow group is also a group of 4 min terms made between adjacent cells of the 4-variable k-maps.
- The green group is a group of 4 min terms made in the left 4-variable k-map. The blue group is of 2 min-terms made in the right 4-variable k-map because there are no common adjacent cells in the other k-map.

- Green color group of 4 min term will produce the term $\bar{A}\bar{C}\bar{E}$. The individual 4-variable K-map will produce $\bar{C}\bar{E}$ as they are not changing in the group but variable A should also be taken into account because this individual 4-variable k-map is being represented by $\bar{A}$.
- The red color group will produce $\bar{C}\bar{D}\bar{E}$. This group is made between both K-maps which means variable A changes and in individual K-map, B changes so these both variables will be eliminated from the term. Only $\bar{C}\bar{D}\bar{E}$ remains unchanged in this group.
- The yellow group will produce $\bar{B}CE$ because these literals are not changing in this group.
- Blue group of 2 min terms will produce the term ABDE as they remain unchanged in this group.
- The simplified expression will be the sum of these 4 terms, which is given below:
- **$F = \bar{A}\bar{C}\bar{E} + \bar{C}\bar{D}\bar{E} + \bar{B}CE + ABDE$**

Example 2 with 5 variable K-map

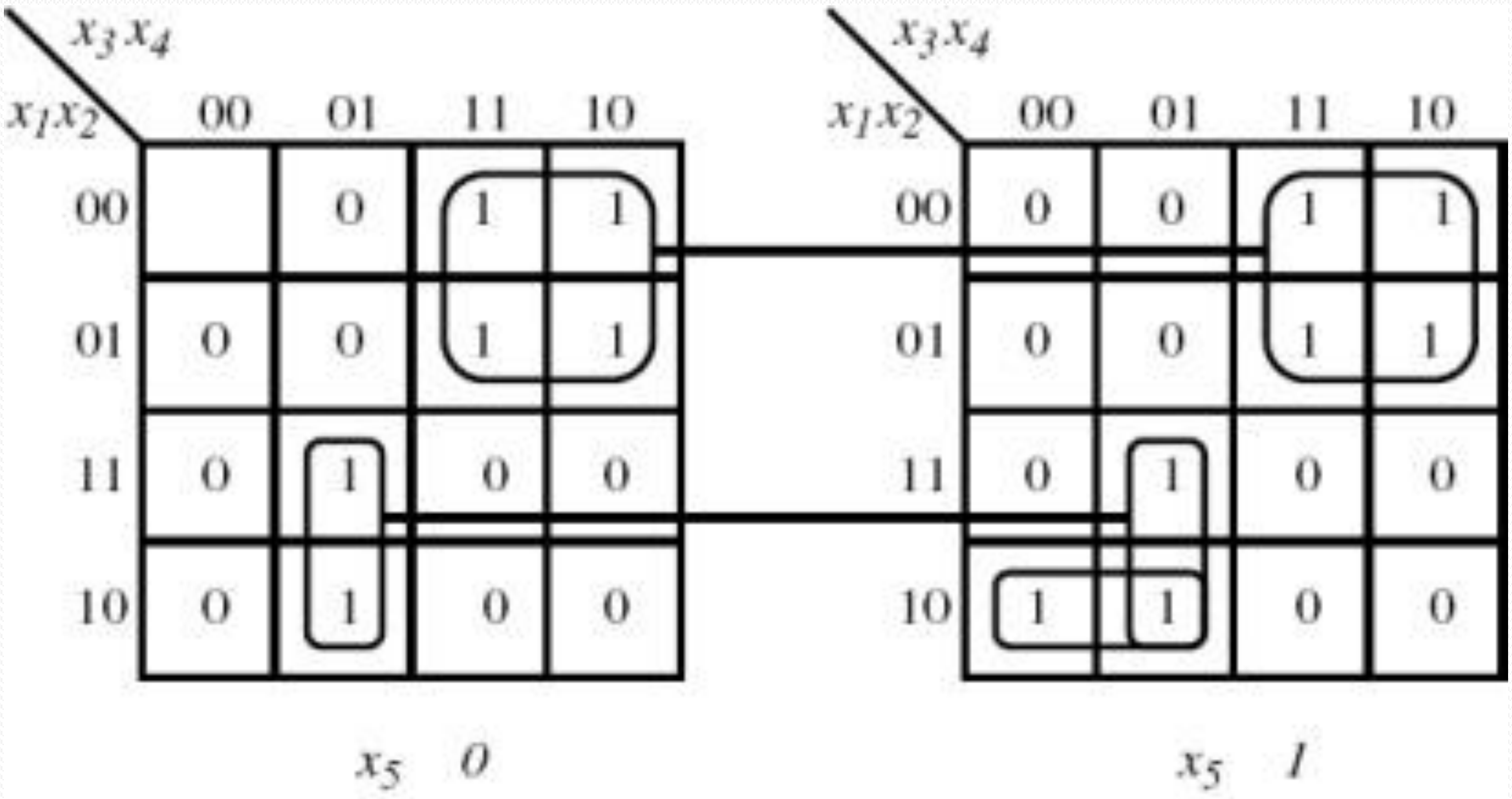


$F = D'E' + B'C'D' + BCE + A'BC'E' + ACDE$

Figure 5-28:

Other Forms of Five-Variable Karnaugh Maps

Example 3 with 5 variable K-map



$$f = \bar{x}_1x_3 + x_1\bar{x}_3x_4 + x_1\bar{x}_2\bar{x}_3x_5$$